

Interpretable Rotor-Bar Fault Diagnosis of Squirrel-Cage Induction Motors Based on Polynomial Temporal Convolutional Network

SUPPLEMENTARY MATERIAL

This Supplementary Material documents the implementation parameters and model configurations used in the revised manuscript. Experimental results, statistical analyses, interpretability discussions, robustness evaluations, and edge-deployment measurement protocols are reported directly in the main manuscript and are not repeated here. The supplement therefore focuses on the configuration details required to reproduce the proposed PTCN and the corresponding comparison experiments.

The implementation information is organized into three tables. Table S-I summarizes the training settings for the Diagonal, Full-shared, and Full-tuned polynomial implementations. It distinguishes the shared-setting comparison, which retains the initialization and structural-regularization settings of the diagonal implementation, from the validation-selected Full-tuned configuration. Table S-II provides the

principal hyperparameters of the proposed PTCN, including the input configuration, polynomial order, PolyConv settings, regularization coefficients, temporal-convolution parameters, and attention settings. Table S-III reports the architectures and architecture-specific hyperparameters of the comparison models used in the experimental evaluation.

Unless otherwise stated, all experiments follow the common training protocol described in the main manuscript: one training epoch corresponding to 655 mini-batch parameter updates, with no early stopping or validation-based best-checkpoint selection. The parameters obtained after the final mini-batch update are used for evaluation. Data preprocessing, complete-recording-wise dataset partitioning, segmentation, and evaluation procedures are consistent with those described in the main manuscript and are therefore not repeated in detail here.

TABLE S-I
TRAINING, REGULARIZATION, AND TUNING SETTINGS OF DIAGONAL AND FULL POLYNOMIAL MODELS

Setting	Diagonal	Full-shared	Full-tuned
Polynomial form	Diagonal	Full local	Full local
Initialization	Fourth-order-priority	Fourth-order-priority	Equal-scale
Optimizer	Adam	Adam	Adam
Learning rate	0.001	0.001	0.001
Batch size	32	32	32
Early stopping	None	None	None
Orthogonality weight	0.02	0.02	0
Energy weight	0.10	0.10	0
Smoothness weight	0.001	0.001	0
Classifier L2	0.1	0.1	0.1
Dropout	0.5	0.5	0.5
Selection criterion	Fixed proposed setting	Fixed shared setting	Validation performance

TABLE S-II.
HYPERPARAMETER SETTINGS OF THE PROPOSED PTCN

Category	Hyperparameter	Value	Category	Hyperparameter	Value
Training	Input size	256×1	Training	Optimizer	Adam
	Number of classes	7		Learning rate	0.001
	Iteration	655		Loss function	Categorical cross-entropy
	Batch size	32		Random seed	42
PolyConv	Maximum order	4	PolyConv	Padding	Same
	Polynomial type	Diagonal		activation	ReLU
	Number of filters	32		Kernel size / stride	4 / 4
Constraints	λ_{orth}	0.02	Downstream Network	TCN dilation rates	1, 2, 4
	λ_{energy}	0.10		TCN kernel size	4
	λ_{smooth}	0.001		Classifier hidden units	128
Downstream	Classifier L2	0.1	Downstream	Attention heads	8
Network	Dropout	0.5	Network	Attention key dimension	16

TABLE S-III.
ARCHITECTURAL CONFIGURATIONS OF THE COMPARISON MODELS

Method	Complete Architecture and Hyperparameter Settings	
TCN	Architecture	Conv1D (16, k=64, s=8) → BN → ReLU → MaxPool (2) → Conv1D (32, k=3) → BN → ReLU → Conv1D (64, k=3) → BN → ReLU → Conv1D (128, k=3) → BN → ReLU → 4 × TCN residual block (dilations 1, 2, 4, 8; filters 64; kernel 3; causal padding; dropout 0.2) → GAP → Dense (128, ReLU) → Dropout (0.3) → Dense (7, Softmax)
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.
CNN	Architecture	Conv1D (64, k=8) → BN → ReLU → MaxPool(2) → Conv1D (64, k=5) → BN → ReLU → MaxPool(2) → Conv1D (64, k=3) → BN → ReLU → MaxPool(2) → GAP → Dense (128, ReLU) → Dropout (0.3) → Dense (7, Softmax).
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.
DCMS	Architecture	Stem Conv1D (16, k=7) → BN → ReLU → Dual-channel multi-scale blocks (Channel A: k ∈ {3,5,7,9}, f=32; Channel B: k ∈ {13,15,17,19}, f=32) → MaxPool (2) per channel → Second MS blocks → Concatenate → Conv1D (128, k=1) → BN → ReLU → GAP → Dense (128, ReLU) → Dropout (0.3) → Dense (7, Softmax).
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.
RMFF	Architecture	5 parallel input branches Conv1D (2, k=1...5) → Concat → 3 Inception-style residual multi-scale fusion blocks (1×1 / 3×3 / 5×5 / MaxPool branches, residual connection) with MaxPool(2) + BN between blocks → Flatten → Dense (256, ReLU) → Dropout (0.3) → Dense (7, Softmax).
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.
LSTM	Architecture	LSTM (128, return_sequences=True) → BN → LSTM (64) → Dense (128, ReLU) → Dropout (0.3) → Dense (7, Softmax).
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.
V-BiLSTM	Architecture	AdaptiveMedianFilter1D(window=5) → WaveletVolterraExpansion (db6-like 3-level decomposition + third-order Volterra terms) → KroneckerFeaturePyramid (dilated convolutions at rates 2, 4, 6) → Conv1D (64, k=5) → BN → ReLU → MaxPool(2) → Conv1D(64, k=3) → BN → ReLU → Conv1D (128, k=3) → BN → ReLU → GAP → Dense (128, ReLU) → Dropout (0.5) → Dense (7, Softmax)
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.
ROAT	Architecture	ROATEnvelopeSpectrum (Hilbert orthogonal axis → pointwise rectified envelope → real FFT magnitude spectrum → log1p) → BN → Conv1D (32, k=5) → BN → ReLU → MaxPool(2) → Conv1D (64, k=3) → BN → ReLU → MaxPool(2) → Conv1D (64, k=3) → BN → ReLU → GAP → Dense (64, ReLU) → Dropout (0.3) → Dense (7, Softmax).
	Settings	Adam (lr = 1×10^{-3}), categorical cross-entropy, batch size 32, 1 epoch.